SIMATS ENGINEERING

SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

CSA37 – SOFTWARE TESTING

EXPERIMENT 15

Test Plan: Inventory Management System

AIM:

To prepare a comprehensive test plan for the Inventory Management System by defining the testing objectives, scope, functional requirements, testing strategies, test environment, execution procedures, entry and exit criteria, roles, responsibilities, and risk mitigation techniques to ensure the system functions accurately, reliably, and securely.

1. Introduction

This test plan defines testing activities for an Inventory Management System used to maintain products, stock levels, suppliers, purchase orders, sales, stock movements, and inventory reports.

2. Objectives

- Verify stock quantities and inventory transactions are accurate.
- Ensure stock updates correctly after receiving, issuing, selling, or returning items.
- Check validation, role permissions, and report accuracy.
- Find defects before release.

3. Test Scope

In scope: login and roles, product/category management, supplier management, stock receipt, stock issue, stock adjustment, reorder alerts, search/filter, purchase orders, transaction history, and reports.

Out of scope: external accounting or shipping systems unless integrated for testing, physical warehouse equipment, and production migration unless included.

4. Functional Requirements

Module	**Requirement**
Authentication	Authorized users can log in and access permitted features.
Product catalogue	Products have unique identifiers and valid details.
Stock receipt	Received quantities increase stock correctly.
Stock issue/sale	Issued quantities reduce stock and follow stock rules.
Adjustment/returns	Authorized changes update stock with an audit trail.
Reorder alert	Alerts appear at configured reorder levels.
Reports	Stock and transaction reports reflect current data.

5. Test Design Techniques

- Equivalence partitioning for valid/invalid product codes, quantities, and supplier details.
- Boundary value analysis for zero, one, maximum quantity, and reorder threshold.
- Decision tables for stock issue permission, available quantity, and approval status.
- State transition testing for purchase order and stock transaction states.
- Use-case testing for receiving stock, issuing stock, correcting errors, and generating reports.

6. Testing Strategies and Methodologies

Testing Type	**Purpose**
Unit testing	Check individual calculations and validation rules.
Integration testing	Verify product, supplier, purchase order, and stock ledger interactions.
System testing	Test complete inventory workflows.
Regression testing	Ensure fixes do not break stock calculations or reports.
Performance testing	Check search and stock updates under expected data volume and users.
Security testing	Verify roles, sessions, and protection against unauthorized stock changes.
Usability testing	Check forms, alerts, and reports for clarity.

7. Test Environment and Data

Use a dedicated test environment, test database, supported browser(s), role-based accounts, and synthetic product/supplier data. Include in-stock, low-stock, zero-stock, and discontinued products. Avoid real production credentials.

8. Test Execution Strategy

- Review requirements and agree on stock and reorder rules.
- Prepare scenarios, cases, and test data.
- Run smoke tests on each build.
- Execute module tests, then integration and end-to-end workflows.
- Test invalid and boundary quantities and concurrent updates where supported.
- Record evidence and log defects with severity and reproducible steps.
- Retest fixes and execute regression tests.
- Publish a final test summary and outstanding-risk list.

9. Entry Criteria

- Requirements and acceptance rules are approved.
- Test build and environment are available.
- Test accounts and data are prepared.

- Critical interfaces required for testing are available.

10. Test Completion Criteria

- All planned critical and high-priority test cases are executed.
- Every case has Pass, Fail, Blocked, or Not Run status recorded.
- No open critical defects remain; high-priority defects are fixed or formally accepted.
- Key inventory workflows and regression tests pass.
- Test summary and known limitations are reviewed by stakeholders.

11. Deliverables

- Test plan, scenarios, and detailed test cases.
- Test data and execution evidence.
- Defect log and retest results.
- Requirements traceability matrix.
- Final test summary report.

12. Roles and Responsibilities

Role	**Responsibility**
Test lead	Plan, coordinate, and report testing.
Tester	Prepare and execute cases and record defects.
Developer	Fix defects and support investigation.
Inventory owner	Confirm business rules and approve residual risks.

13. Risks and Mitigation

Risk	**Mitigation**
Incorrect stock rules	Confirm returns, adjustments, and negative-stock rules.
Concurrent updates cause mismatch	Test simultaneous transactions and database consistency.
Incomplete supplier/product data	Prepare representative test datasets.
Late changes	Use impact analysis and regression testing.

14. Conclusion

This test plan defines the scope, test design techniques, execution strategy, and completion criteria required to evaluate the Inventory Management System and support a reliable release.